

JoTouch Model Comparison Results

- Training budget (identical within each pipeline): 12 epochs, batch 64, lr=0.001
- Alternative-models pipeline: plain Adam only — no OneCycleLR, no early stopping, no gradient clipping
- JoTouch pipeline (ours): AdamW + OneCycleLR + early stopping (patience 4) + gradient clipping
- Device: cpu

Dataset: Zenodo MDPI FMG (17 subjects, classification + regression)

- Source: Zenodo 6632020 (Zakia & Menon, MDPI Data 2022) - real FMG, first 8 of 32 channels, biaxial-stage motions
- Windows: 12,500 | Subjects: 17 | Classes: 5 (1D_X, 1D_Y, 2D_DG, 2D_DM, 2D_SQ)

Classification

Model	Params	Size (KB)	Random-split Acc	Cross-Subject Acc (LOSO mean±std)	Generalization Gap	Latency (ms)
Linear (oldest)	805	3.1	47.24%	30.09% ± 3.87%	17.15 pts	0.03
Standard CNN (older alternative)	11,493	44.9	75.84%	34.91% ± 3.08%	40.93 pts	0.27
LSTM + MobileNetV1 (older alternative)	42,181	164.8	92.04%	34.47% ± 4.32%	57.57 pts	1.19
JoTouch (ours)	65,797	257.0	96.76%	35.38% ± 9.64%	61.38 pts	1.48
JoTouch-V3 (upgrade candidate)	70,053	273.6	96.80%	35.71% ± 7.31%	61.09 pts	1.50

Per-class accuracy (cross-subject, aggregated over folds)

Class	Linear (oldest)	Standard CNN (older alternative)	LSTM + MobileNetV1 (older alternative)	JoTouch (ours)	JoTouch-V3 (upgrade candidate)
1D_X	41.1%	48.8%	48.6%	47.4%	46.3%
1D_Y	47.2%	37.4%	35.9%	36.4%	42.5%
2D_DG	23.6%	41.2%	42.7%	38.5%	38.0%
2D_DM	33.2%	36.8%	33.6%	32.8%	29.7%
2D_SQ	4.7%	13.4%	16.0%	22.8%	23.2%

Regression (continuous targets)

Model	Random-split MAE	Random-split R²	Cross-Subject MAE (LOSO mean±std)	Cross-Subject R²
Linear (oldest)	0.0722	0.225	0.0999 ± 0.0294	-0.561 ± 0.656
Standard CNN (older alternative)	0.0590	0.447	0.0829 ± 0.0110	-0.134 ± 0.168
LSTM + MobileNetV1 (older alternative)	0.0349	0.808	0.0886 ± 0.0008	-0.303 ± 0.103
JoTouch (ours)	0.0297	0.847	0.0781 ± 0.0088	0.026 ± 0.119
JoTouch-V3 (upgrade candidate)	0.0321	0.823	0.0770 ± 0.0110	0.048 ± 0.167

Dataset: Exoskelebox FMG benchmark (20 subjects, 8 FSR, 6 gestures)

- Source: Exoskelebox FMG benchmark (arXiv:2007.14918) - 8 forearm FSR, 20 subjects, 6 gestures x 3 wrist orientations, ~100Hz after x5 downsample
- Windows: 15,000 | Subjects: 20 | Classes: 6 (closed, extension, flexion, rest, straight, wide)

Classification

Model	Params	Size (KB)	Random-split Acc	Cross-Subject Acc (LOSO mean±std)	Generalization Gap	Latency (ms)
Linear (oldest)	966	3.8	29.10%	19.56% ± 3.52%	9.54 pts	0.06
Standard CNN (older alternative)	11,558	45.1	46.43%	25.20% ± 3.21%	21.23 pts	0.45
LSTM + MobileNetV1 (older alternative)	42,246	165.0	59.03%	24.79% ± 4.12%	34.24 pts	1.97
JoTouch (ours)	65,862	257.3	64.47%	21.23% ± 2.72%	43.24 pts	2.44
JoTouch-V3 (upgrade candidate)	70,118	273.9	63.27%	21.76% ± 2.55%	41.50 pts	1.78

Per-class accuracy (cross-subject, aggregated over folds)

Class	Linear (oldest)	Standard CNN (older alternative)	LSTM + MobileNetV1 (older alternative)	JoTouch (ours)	JoTouch-V3 (upgrade candidate)
closed	37.1%	54.5%	46.8%	22.5%	27.9%
extension	14.7%	39.2%	45.1%	45.2%	40.4%
flexion	17.3%	10.6%	12.0%	13.0%	13.1%
rest	11.8%	16.4%	12.6%	6.5%	6.6%
straight	19.6%	14.1%	9.7%	16.7%	14.8%
wide	18.9%	18.0%	24.8%	25.0%	29.2%

Dataset: PLoS ONE 2025 FMG (27 participants, 4 gestures + load regression)

- Source: PLoS ONE 2025 (Young et al., e0321319) - Zenodo 15420178, 27 participants, 8 FMG ch @100Hz after x20 downsample, 4 gestures x 5 loads; regression target = grasped load (normalized)
- Windows: 10,000 | Subjects: 27 | Classes: 4 (Key, Pinch, Power, Tripod)

Classification

Model	Params	Size (KB)	Random-split Acc	Cross-Subject Acc (LOSO mean±std)	Generalization Gap	Latency (ms)
Linear (oldest)	644	2.5	32.90%	27.62% ± 2.12%	5.28 pts	0.04
Standard CNN (older alternative)	11,428	44.6	39.40%	28.83% ± 1.08%	10.57 pts	0.25
LSTM + MobileNetV1 (older alternative)	42,116	164.5	49.40%	28.46% ± 1.10%	20.94 pts	1.23
JoTouch (ours)	65,732	256.8	50.65%	29.36% ± 1.45%	21.29 pts	1.37
JoTouch-V3 (upgrade candidate)	69,988	273.4	51.10%	28.55% ± 0.44%	22.55 pts	1.73

Per-class accuracy (cross-subject, aggregated over folds)

Class	Linear (oldest)	Standard CNN (older alternative)	LSTM + MobileNetV1 (older alternative)	JoTouch (ours)	JoTouch-V3 (upgrade candidate)
Key	34.2%	49.3%	44.3%	41.7%	40.1%
Pinch	24.5%	26.0%	23.2%	19.4%	18.5%
Power	17.0%	20.7%	21.9%	37.7%	24.3%
Tripod	34.7%	19.4%	24.5%	18.5%	31.2%

Regression (continuous targets)

Model	Random-split MAE	Random-split R ²	Cross-Subject MAE (LOSO mean±std)	Cross-Subject R ²
Linear (oldest)	0.3044	0.009	0.3272 ± 0.0160	-0.260 ± 0.171
Standard CNN (older alternative)	0.2997	0.033	0.3030 ± 0.0044	-0.018 ± 0.031
LSTM + MobileNetV1 (older alternative)	0.2978	0.053	0.3182 ± 0.0124	-0.160 ± 0.131
JoTouch (ours)	0.2985	0.049	0.3058 ± 0.0001	-0.024 ± 0.009
JoTouch-V3 (upgrade candidate)	0.2975	0.048	0.3051 ± 0.0017	-0.018 ± 0.019

MobileNetV3 upgrade evaluation (should JoTouch replace MobileNetV2 with MobileNetV3?)

JoTouch (ours, MobileNetV2 + TCN) vs JoTouch-V3 candidate (MobileNetV3 + TCN), trained with the identical JoTouch pipeline on identical splits:

Dataset	MBV2 Cross-Subject Acc	V3 Cross-Subject Acc	Δ Acc	MBV2 Latency (ms)	V3 Latency (ms)	MBV2 Params	V3 Params
Zenodo MDPI FMG (17 subjects, classification + regression)	35.38% ± 9.64%	35.71% ± 7.31%	+0.32 pts	1.48	1.50	65,797	70,053
Exoskeletonbox FMG benchmark (20 subjects, 8 FSR, 6 gestures)	21.23% ± 2.72%	21.76% ± 2.55%	+0.54 pts	2.44	1.78	65,862	70,118
PLoS ONE 2025 FMG (27 participants, 4 gestures + load regression)	29.36% ± 1.45%	28.55% ± 0.44%	-0.81 pts	1.37	1.73	65,732	69,988

Decision: KEEP MobileNetV2. MobileNetV3 shows no meaningful gain (mean Δ = +0.02 pts across 3 datasets) while adding SE-block parameters and latency — the theoretical ImageNet advantage does not materialize on 8-channel FMG windows, so the simpler V2 stays.

Metrics explained

- Params / Size (KB): number of trainable weights and approximate float32 model size — matters for embedding on microcontrollers (flash/RAM limits).
- Random-split Acc: accuracy on a random 80/20 window split; windows from the same trial can appear in both sets — the optimistic, in-session number.

- Cross-Subject Acc (LOSO): leave-subjects-out cross-validation — each fold excludes whole subjects from training and tests only on them; mean $\pm$ std across folds. The realistic 'new user' number and the key generalization evidence.
- Generalization Gap: random-split accuracy minus cross-subject accuracy; smaller = less overfitting.
- Latency (ms): mean of 100 single-window inference passes after a 10-pass warm-up (same method as test_latency in JoTouch_AI_module.py); must stay well under the 10 ms real-time budget.
- MAE / RMSE: mean absolute / root-mean-square error of the continuous targets (normalized 0–1); lower is better.
- R²: coefficient of determination; 1.0 = perfect prediction, 0 = no better than predicting the mean, negative = worse than the mean.